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• structural understanding • primitive-guided post-training

The Missing Primitive: Diagnosing and Repairing Mathematical Reasoning in Large Language Models

A New Benchmark Reveals that Discovery—Not Execution—Is the Universal Bottleneck to LLM Mathematical Reasoning, and That Targeting It Yields Consistent Gains

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LLMs that routinely solve competition-level mathematics often lack the structural understanding underlying their solutions: given the correct foundational idea—a *Mathematical Primitive*—the same models execute near-perfectly, but independently discovering that primitive is the dominant bottleneck across all frontier architectures tested. Primitive-guided post-training closes the gap.

AT A GLANCE

Problem: LLM mathematical reasoning accuracy does not reveal whether models possess the structural understanding behind their solutions.

Why it matters: Models with identical benchmark scores may rely on brittle heuristics; diagnosing structural gaps enables targeted improvement.

Method: Mathematical Primitive concept plus the Prim benchmark spanning Discovery, Generation, Digestion, and Execution dimensions.

Result: Discovery lags Execution by up to 30 pp; primitive-guided post-training consistently raises both Discovery and overall task accuracy.

Evidence: Evaluated across multiple frontier LLMs on Prim and held-out downstream math benchmarks.

The Big Picture

Large language models have crossed remarkable thresholds in mathematical reasoning: the best contemporary systems solve a substantial fraction of AMC and AIME problems and approach expert performance on graduate-level benchmarks. Yet this aggregate success conceals a potentially critical gap: does solving a problem indicate that the model grasped the foundational idea organizing

the solution, or merely that it reproduced a pattern from training data?

This paper takes a first step toward answering that question by introducing the concept of the *Mathematical Primitive*—a concise, non-procedural description of the core insight that, once identified, makes the detailed execution steps straightforward. For example, a combinatorics problem’s primitive might be “map to a stars-and-bars counting argument” rather than the specific arithmetic that follows from it.

Why Existing Approaches Fall Short

Benchmark saturation: State-of-the-art LLMs now score near ceiling on MATH and AMC; high scores mask vastly different capability profiles.

No structural audit: Existing benchmarks measure correct final answers, not whether the model identified the organizing idea. Two models with identical accuracy may differ dramatically in structural understanding.

Post-training targeting: Standard chain-of-thought fine-tuning and process reward models treat all reasoning steps equally; they do not specifically target primitive discovery where diagnosis reveals the greatest weakness.

Core Mechanism

Mathematical Primitive. A primitive is defined as the single, concise, non-procedural representation of the foundational idea that organizes a solution. It is shorter than a proof sketch, more specific than a problem category, and must identify the key conceptual move.

Prim Benchmark. The paper constructs Prim with four complementary dimensions:

- *Discovery:* Given the problem, identify the primitive.
- *Generation:* Given the primitive, produce a valid solution.
- *Digestion:* Given a complete solution, extract the primitive it relies on.
- *Execution:* Given both problem and primitive, carry out the solution steps.

Technical Formulation

Let $\mathcal{P}$ be the space of primitives and $\mathcal{Q}$ the space of problems. Define four binary capability indicators for a model $\mathcal{M}$ on problem q with ground-truth primitive $p^*(q)$:

$$\text{Disc}(q) = \mathbf{1}[\hat{p}(q) = p^*(q)]$$

$$\text{Exec}(q, p^*) = \mathbf{1}[\hat{s}(q, p^*(q)) \text{ is correct}]$$

The central finding is:

$$\mathbb{E}_q[\text{Exec}(q, p^*(q))] \gg \mathbb{E}_q[\text{Disc}(q)]$$

for all tested models, establishing that **Discovery is the dominant bottleneck**. Primitive-guided post-training minimizes a combined loss:

$$\mathcal{L} = \mathcal{L}_{\text{solve}} + \lambda \mathcal{L}_{\text{discover}}$$

where $\mathcal{L}_{\text{discover}}$ is a cross-entropy term on primitive Discovery, and λ is a balancing hyperparameter.

Learning or Inference Procedure

Evaluation:

1. Run each Prim dimension independently on the same model.
2. Compute per-dimension accuracy and capability profile.
3. Compare profiles across models with similar aggregate task accuracy.

Post-training:

1. Annotate training problems with ground-truth or LLM-generated primitives.
2. Add a primitive Discovery objective alongside the standard solution objective.
3. Fine-tune with combined loss $\mathcal{L}$; evaluate on held-out Prim dimensions and downstream benchmarks.

What the Guarantee Says

No formal convergence bound is given; the Discovery bottleneck is an empirical finding. The central claim is: across all tested frontier LLMs, Execution given a correct primitive substantially exceeds unconditioned Discovery, establishing the latter as universally limiting. Post-training improvement transfers to held-out benchmarks not seen during fine-tuning.

Experimental Findings

- **Discovery gap:** Execution given the correct primitive exceeds unconditioned Discovery by up to 30 percentage points across all frontier LLMs.
- **Hidden differences:** Two models with identical MATH accuracy can differ by over 20 pp on the Discovery dimension alone.
- **Latent execution capacity:** Providing correct primitives to models that cannot discover them independently unlocks near-full solution accuracy.
- **Post-training gains:** Primitive-guided fine-tuning consistently improves Discovery accuracy and overall task performance, with gains transferring to held-out sets.

Ablations and Interpretation

Primitive granularity: Coarser primitives yield lower Execution improvement; fine-grained primitives that specify the exact conceptual move are most effective.

Imperfect primitives: Using LLM-generated rather than ground-truth primitives reduces but does not eliminate the Execution advantage—even imperfect discovery guidance helps.

Digestion vs. Discovery: High Digestion accuracy does not predict high Discovery accuracy, establishing that recognizing a primitive in a given solution and independently finding it from a problem statement are distinct capabilities.

Reference

Shuo Xing et al. **The Missing Primitive: Diagnosing and Repairing Mathematical Reasoning in Large Language Models**. arXiv:2610.02191, October 2026.

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Keywords: diffusion models • inference-time scaling • Feynman-Kac correction • molecular free energy

Error-Corrected Inference-Time Scaling for Imperfect Diffusion Models

The Energy-based Feynman-Kac Corrector Removes Systematic Model Bias at Inference Time, Achieving Accurate Molecular Free-Energy Profiles Where Standard Particle Methods Fail

By Zuokai Wen, Louis Grenioux, Weinan E & Jiequn Han (Princeton University; Flatiron Institute, Simons Foundation)

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Inference-time scaling for diffusion models—running more particles through sequential Monte Carlo—reduces variance but cannot remove the systematic bias introduced by an imperfect pretrained score function. The Energy-based Feynman-Kac Corrector (EBFKC) derives dynamics that track a prescribed probability path exactly in the population limit regardless of model error, and demonstrates accurate molecular free-energy profiling on alanine dipeptide and tetrapeptide where standard methods fail.

AT A GLANCE

Problem: Standard inference-time scaling inherits the systematic bias of an imperfect diffusion model; more particles reduce variance but not bias.

Why it matters: Scientific applications (molecule generation, free-energy computation) require accurate tail distributions, not just good mode coverage.

Method: Feynman-Kac potential derived from an energy-based reference corrects the imperfect score on the fly; approximated via sequential Monte Carlo with variance-controlling guidance.

Result: Closely matches ground-truth free-energy surfaces on alanine dipeptide and tetrapeptide; standard SMC baselines retain several $k_B T$ of error.

Evidence: Gaussian mixture models, particle systems, two alanine peptides under annealing and reward tilting.

The Big Picture

Inference-time scaling has emerged as a key technique for improving diffusion model quality: by running a sequential Monte Carlo (SMC) procedure with many particles, one can steer generation toward a target distribution or reward signal without retraining the model. Applications range from image generation with aesthetic rewards to molecular

conformation sampling with energy-based targets.

All existing approaches, however, share a silent assumption: that the pretrained diffusion model’s score function is exact. In practice, score functions are trained on finite data with finite compute and carry systematic approximation error. Standard SMC reduces the statistical error of the particle approximation but cannot remove this model-level bias: every particle evolves under the same imperfect dynamics, so the distribution they jointly represent converges to the wrong target.

Why Existing Approaches Fall Short

Monte Carlo only reduces variance, not bias. Adding more particles in standard SMC reduces $O(1/\sqrt{N})$ variance, leaving a fixed systematic error proportional to the score function mismatch $\epsilon_t = \hat{s}_\theta - \nabla \log p_t$.

Path tracking error. Imperfect scores cause particles to drift off the prescribed probability path, accumulating errors that compound over diffusion timesteps.

Reward tilting. Reward-guided diffusion reweights samples assuming a correct model; systematic score error introduces additional bias into the tilted distribution that no reweighting scheme can correct post-hoc.

Core Mechanism

Energy-based Feynman-Kac Corrector (EBFKC). Given a reference energy $E_t(\mathbf{x})$ defining the desired path $\pi_t \propto e^{-E_t(\mathbf{x})}$, the paper derives a Feynman-Kac potential G_t that, when incorporated as a multiplicative weight in the SMC procedure, ensures the particle ensemble tracks the prescribed path $\{p_t\}$ exactly in the continuous-time population limit, regardless of the score function error ϵ_t .

The key insight is that the Feynman-Kac formula from stochastic analysis provides a closed-form expression for the correction term that neutralizes the effect of ϵ_t without requiring access to the true score function—only the reference energy is needed.

Technical Formulation

Let $\hat{s}_\theta(\mathbf{x}, t)$ be the imperfect score with error $\epsilon_t(\mathbf{x}) = \hat{s}_\theta - \nabla \log p_t$. Standard SMC propagates under:

$$d\mathbf{x}_t = \hat{s}_\theta(\mathbf{x}_t, t) dt + \sqrt{2} dW_t$$

and tracks p_t only when $\epsilon_t = 0$.

EBFKC derives the Feynman-Kac potential:

$$G_t(\mathbf{x}) = \exp\left(-\int_0^t \langle \epsilon_\tau(\mathbf{x}_\tau), \nabla \log p_\tau(\mathbf{x}_\tau) \rangle d\tau\right)$$

The corrected particle system with weights proportional to G_t converges to p_t in the population limit. A variance-

controlling guidance term:

$$\hat{s}_\theta^{\text{VC}}(\mathbf{x}, t) = \hat{s}_\theta(\mathbf{x}, t) - \gamma \nabla_{\mathbf{x}} \log w_t(\mathbf{x})$$

prevents particle collapse by penalizing low-weight regions, where $\gamma > 0$ is a guidance strength parameter.

Learning or Inference Procedure

EBFKC requires no additional training—it operates on a frozen pretrained diffusion model.

1. **Specify** reference energy E_t (free energy functional or reward model).
2. **Initialize** N particles $\{\mathbf{x}_T^{(i)}\}_{i=1}^N$ from the prior p_T .
3. **Propagate** with $\hat{s}_\theta$ plus variance-controlling guidance, accumulating Feynman-Kac weights $G_t^{(i)}$ at each step.
4. **Resample** when the effective sample size $N_{\text{eff}} = (\sum_i w_i^2)^{-1}$ drops below $N/2$.
5. **Return** the weighted ensemble $\{(\mathbf{x}_0^{(i)}, w_0^{(i)})\}$.

The overhead relative to standard SMC is one energy evaluation per particle per step.

What the Guarantee Says

In the continuous-time population limit ($N \rightarrow \infty$, $\Delta t \rightarrow 0$), EBFKC tracks the prescribed path $\{p_t\}$ exactly, independent of ϵ_t . For finite N , the error scales as $O(1/\sqrt{N})$ in particle count and $O(\Delta t)$ in discretization step size. Crucially, unlike standard SMC, the $N \rightarrow \infty$ limit converges to the correct target rather than a biased surrogate.

Experimental Findings

- **Gaussian mixture models:** EBFKC recovers all modes of a multi-modal 2D target; standard SMC with imperfect model overweights dominant modes.
- **Particle systems:** EBFKC accurately recovers equilibrium distributions for interacting particles; standard baselines show persistent density errors.
- **Alanine dipeptide:** EBFKC matches the ground-truth Ramachandran free-energy surface within statistical uncertainty; standard SMC exhibits errors of several $k_B T$ in high-barrier regions.
- **Alanine tetrapeptide:** EBFKC scales to the larger system and maintains accurate free-energy profiles under both annealing and reward-tilting regimes.

Ablations and Interpretation

Particle count: EBFKC converges faster to the correct target than standard SMC as N increases; at low N , variance-controlling guidance is essential.

Model imperfection level: As score error ϵ_t is arti-

ficially increased, standard SMC degrades rapidly while EBFKC remains accurate.

Variance-controlling guidance: Removing this term causes particle collapse at long horizons, confirming its necessity.

Feynman-Kac vs. naive reweighting: Simple importance weighting without the Feynman-Kac path correction tracks a biased distribution even with large N , underscoring that the correction must operate on the dynamics, not just the terminal weights.

Reference

Zuokai Wen, Louis Grenioux, Weinan E, Jiequn Han. **Error-Corrected Inference-Time Scaling for Imperfect Diffusion Models**. arXiv:2610.01933, October 2026.

<https://arxiv.org/abs/2610.01933>

Keywords: time series foundation models • zero-shot forecasting • benchmarking • temporal inductive bias

Foundations without Fundamentals: Zero-Shot Blind Spots in Time Series Foundation Models

Chronos-2, Moirai, and Toto Fail Unit Tests for Monotonic Trends, Periodic Signals, and Leading Indicators That Classical Models Pass Trivially

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Despite strong performance on broad benchmarks, prominent Time Series Foundation Models fail surprisingly basic unit tests—monotonic trends, periodic signals, and leading-indicator exploitation—that simple classical models handle trivially, revealing a gap between pre-training scale and fundamental temporal inductive biases that persists on real-world sensor data.

AT A GLANCE

Problem: TSFMs may lack inductive biases for basic temporal patterns despite strong broad-benchmark performance.

Why it matters: Monotonic trends, periodicity, and causal leading indicators are ubiquitous in industrial, scientific, and healthcare time series.

Method: SimpleTimeBench—a diagnostic suite of unit tests for temporal primitives (trend, periodicity, leading indicators) at multiple difficulty levels.

Result: Chronos-2, Moirai, and Toto frequently fail zero-shot unit tests; ARIMA/Theta match or exceed them on basic primitives.

Evidence: Synthetic controlled series plus real-world building energy meters and meteorological stations.

The Big Picture

Time Series Foundation Models (TSFMs) trained on massive corpora—Chronos-2 on billions of time points, Moirai on diverse multi-domain data—have achieved impressive zero-shot performance on established benchmarks such as Monash and ETT. The natural inference is that these models have internalized general temporal structure, including basic primitives like monotonic trends, cyclical seasonality, and causal leading indicators.

This paper challenges that inference by constructing Sim-

pleTimeBench: a diagnostic benchmark of unit tests for temporal primitives where any well-calibrated forecaster should produce near-perfect zero-shot results. If a model with billions of training examples fails these trivial tests, the gap between pre-training scale and genuine temporal reasoning is larger than aggregate benchmarks suggest.

Why Existing Approaches Fall Short

Conflation problem: Aggregate benchmark performance combines hundreds of heterogeneous series; strong performance on diverse data does not guarantee competence on any individual temporal primitive.

No unit tests: There is no existing benchmark analogous to software unit tests that verifies whether a model handles basic temporal building blocks correctly.

Invisible covariate gaps: Existing benchmarks rarely test whether models exploit available exogenous covariates (leading indicators), so systematic failure to use causal information goes undetected.

Core Mechanism

SimpleTimeBench comprises three test families:

- *Monotonic trend:* Univariate series $x_t = \alpha t + \sigma \epsilon_t$ with varying slope α and noise σ . The oracle continues the trend; reversals or early flattening constitute failures.
- *Periodic signals:* Series with known periods (daily/weekly/annual) and controlled amplitudes. A model with temporal periodicity inductive bias should closely track cycles.
- *Leading indicators:* Multivariate series where channel x_{t-k} Granger-causes y_t with a known lag k , providing complete information for an accurate forecast.

Each family spans multiple noise levels, slopes, periods, and lag lengths to characterize sensitivity rather than produce a binary pass/fail verdict.

Technical Formulation

For the monotonic trend test at noise level σ and slope α :

$$x_t = \alpha t + \sigma \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1)$$

The oracle forecast for horizon h is $\hat{x}_{T+h} = \alpha(T + h)$, achieving $\text{MASE} \approx 1$. A model *passes* if $\text{MASE}_{\text{model}}/\text{MASE}_{\text{oracle}} < \delta$ for threshold $\delta = 2.0$.

For the leading indicator test with lag k :

$$y_t = \phi(x_{t-k}) + \sigma \epsilon_t$$

The oracle MSE is σ^2 . The leading-indicator utilization rate ρ is:

$$\rho = 1 - \frac{\text{MSE}_{\text{model}} - \sigma^2}{\text{MSE}_{\text{no-cov}} - \sigma^2}$$

where $\text{MSE}_{\text{no-cov}}$ is the MSE of a univariate model that ignores x . A ρ near 1 indicates full utilization; TSFMs average $\rho < 0.4$.

Learning or Inference Procedure

SimpleTimeBench is evaluation-only. The procedure:

1. Generate controlled synthetic series at multiple difficulty levels.
2. Run each TSFM (Chronos-2, Moirai, Toto) in zero-shot mode with the model’s default context window.
3. Compute MASE, sMAPE, and ρ for each test family.
4. Compare against oracle forecasters and classical baselines (ARIMA, Theta, ARIMAX).
5. Validate on real-world sensor datasets containing the tested primitives.

What the Guarantee Says

SimpleTimeBench provides a falsifiability guarantee: failure on any test family certifies the existence of natural inputs (with no distributional shift) for which the model underperforms a trivially simple oracle. This is a constructive, worst-case characterization—not a claim about average performance.

Experimental Findings

- **Trend reversal:** Chronos-2 reverses an increasing trend in $\approx 18\%$ of test cases at low noise; Moirai flattens trends early in $\approx 23\%$.
- **Periodic signals:** All three models exhibit MASE ratios above 2.0 relative to the oracle in $> 30\%$ of annual-period instances.
- **Leading indicators:** Average utilization rate $\rho < 0.40$ for all three TSFMs; ARIMAX achieves $\rho > 0.80$ on the same series.
- **Real-world sensors:** On building energy meters, TSFMs consistently underperform a tuned Fourier decomposition model on strongly periodic signals.
- **Classical model comparison:** ARIMA and Theta match or exceed TSFMs on monotonic trend and leading indicator tests despite being far simpler.

Ablations and Interpretation

Context window: Longer context modestly improves periodic performance (5–10% MASE reduction) but does not eliminate failures.

Model size: Larger model variants do not substantially improve SimpleTimeBench pass rates—the blind spots are not simply a parameter-count issue.

Fine-tuning: A small amount of domain fine-tuning on

primitive-containing examples raises pass rates to $> 85\%$, confirming the biases are learnable but absent in zero-shot pre-trained form.

Reference

Nafiseh Ghoroghchian, Haipeng Zhang, Shuyi Han, Alex Labach, George Stein. **Foundations without Fundamentals: Zero-Shot Blind Spots in Time Series FMs.** arXiv:2610.02058, October 2026.
<https://arxiv.org/abs/2610.02058>

Keywords: mechanistic interpretability • non-linear manifolds • concept alignment • LLM representations

Beyond Linear Concepts: Discovering and Aligning Non-Linear Concept Manifolds in Large Language Models

Modelling Concepts as Low-Dimensional Riemannian Manifolds Reveals Consistent Block Structures and a Syntax-to-Semantics Transition Universal Across Architectures

By Tido Specht, Elias Benedict Krey, Nils Neukirch & Nils Strodthoff (University of Oldenburg; DFKI)

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Adapting Non-Linear Multi-Dimensional Concept Discovery (NLMCD) from computer vision to token-level LLM activations and introducing the Concept-Based Alignment (CBA) score, this work reveals that concept organization in LLMs exhibits consistent block structures across layers and architectures, with representations staying syntax-dominated through 70% of the network before transitioning to mixed syntactic-semantic structure in late layers—patterns invisible to linear methods.

AT A GLANCE

Problem: Existing MI methods assume linear concept representations, missing the non-linear geometric structure of LLM activations.

Why it matters: Linear methods may produce incomplete or misleading interpretations of how concepts are organized in large models.

Method: NLMCD fits low-dimensional manifolds to token activations; CBA score (generalized Rand index) compares manifolds across layers and models without requiring explicit feature matching.

Result: Consistent block structures at intermediate layers; syntax-dominated through 70% of network then mixed syntactic-semantic; CBA is 15–30% more sensitive than PCA or CKA.

Evidence: Five LLM architectures (Llama, Mistral, Falcon, and others) on Wikipedia + code corpora.

The Big Picture

Mechanistic interpretability seeks to decode the internal representations of large language models—how concepts are organized, transformed, and composed across layers. Dominant methods (probes, sparse autoencoders, dictionary learning) rest on a shared linearity assumption: each

concept occupies a direction in activation space, and its presence is measured by a linear classifier or inner product.

This assumption is increasingly challenged by empirical evidence showing curved, non-linearly distributed activation clusters. If concepts genuinely occupy low-dimensional manifolds rather than directions, linear methods will systematically miss or mischaracterize them. This paper takes the manifold hypothesis seriously and develops tools to act on it, revealing organizational principles in LLMs that linear baselines cannot detect.

Why Existing Approaches Fall Short

Linear probes measure concept presence as a dot product with a learned direction, entirely missing curved or non-linearly distributed concept regions.

Sparse autoencoders (SAEs) decompose activations into sparse linear combinations; each dictionary feature is a direction, and compositions are linear sums. Non-linear manifold structure is encoded only incidentally.

CKA and PCA-based comparisons compare linear subspaces across layers using canonical angles; they are insensitive to non-linear geometry and cannot detect manifold-level structural differences.

No cross-model alignment: Existing methods lack a rigorous way to compare concept organization across different architectures without explicit feature matching, which assumes a known bijection between concepts—not generally available.

Core Mechanism

NLMCD on LLM Activations. For each concept category c (syntactic POS tag, dependency role, semantic entity type) and each layer l , the paper collects token activation sets $\mathcal{H}_c^l$ and fits a low-dimensional Riemannian manifold $\mathcal{M}_c^l$ using a neural network atlas.

Concept-Based Alignment (CBA) Score. CBA measures whether two manifolds partition the same token set into the same neighborhoods, without requiring explicit feature matching:

$$\text{CBA}(\mathcal{M}_c^l, \mathcal{M}_c^{l'}) = \text{RandIndex}(G_c^l, G_c^{l'})$$

where G_c^l is a neighborhood graph on tokens $\mathcal{T}_c$ derived from geodesic distances on $\mathcal{M}_c^l$. CBA is 1.0 when both manifolds partition the token space identically and 0.5 at chance.

Technical Formulation

Let $\mathbf{h}_l^{(t)} \in \mathbb{R}^d$ be the activation of token t at layer l . The NLMCD atlas $\phi_c^l : \mathbb{R}^k \rightarrow \mathbb{R}^d$ (with $k \ll d$) is fitted by:

$$\min_{\phi} \sum_{t \in \mathcal{T}_c} \|\mathbf{h}_l^{(t)} - \phi(z^{(t)})\|^2 + \lambda \text{Reg}(\phi)$$

where $z^{(t)} \in \mathbb{R}^k$ are latent coordinates learned jointly, and $\text{Reg}(\phi)$ is a smoothness regularizer.

The CBA between concept c at layers l and l' uses the generalized Rand index:

$$\text{CBA} = \frac{|\{(i, j) : \text{near in } G_c^l \Leftrightarrow \text{near in } G_c^{l'}\}|}{|\mathcal{T}_c|^2}$$

All-pairs CBA matrices across layers $\{1, \dots, L\}$ reveal which layer intervals share similar concept geometry.

Learning or Inference Procedure

Manifold fitting (offline):

1. Run the LLM on a large corpus ($\approx 10\text{M}$ tokens).
2. Extract activations $\{\mathbf{h}_l^{(t)}\}$ for all tokens and layers.
3. For each concept category c and layer l , fit NLMCD atlas ϕ_c^l using stochastic gradient descent.

CBA computation (post-fitting):

1. Compute pairwise geodesic distances on each fitted manifold.
2. Build neighborhood graphs G_c^l and compute Rand index between all (layer, model) pairs.
3. Visualize as CBA alignment matrices to identify block structures.

What the Guarantee Says

NLMCD provides a consistent manifold estimate as sample size grows, under standard regularity conditions on manifold geometry. CBA is a metric on the space of neighborhood partitions. The main empirical claim—cross-architecture consistency of block structures—is validated by replicating CBA patterns across five model families including architectures trained on different data regimes.

Experimental Findings

- **Block structures:** CBA matrices show consistent high-alignment blocks at intermediate layers ($\approx$ layers 10–20 in a 32-layer model).
- **Syntax-to-semantics:** Syntactic concepts dominate CBA structure through $\approx 70\%$ of network depth; semantic concepts become prominent in the final 30% of layers across all tested models.
- **Cross-architecture:** Block structures are consistent across Llama, Mistral, Falcon and two additional

families at equivalent depth fractions.

- **CBA sensitivity:** CBA is 15–30% more sensitive than PCA-based or CKA-based baselines, as measured by detection of synthetic perturbations.
- **Non-linearity:** Linear probes in place of NLMCD miss the block transitions entirely, confirming that non-linear geometry carries genuine structural information.

Ablations and Interpretation

Manifold dimension k : Values $k \in \{2, 4, 8\}$ all produce similar CBA patterns; $k = 4$ offers the best fidelity-cost trade-off.

Corpus composition: Code-heavy vs. text-heavy corpora show similar syntactic block patterns but differ in semantic concept structure of late layers, aligned with known training data differences.

Tokenization: Character-level tokens show less pronounced block structure than subword tokens, suggesting the patterns are tightly coupled to the tokenization scheme.

Reference

Tido Specht, Elias Benedict Krey, Nils Neukirch, Nils Strödtz. **Beyond Linear Concepts: Discovering and Aligning Non-Linear Concept Manifolds in Large Language Models.** arXiv:2610.01821, October 2026.

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Keywords: memory-efficient fine-tuning • operator-norm optimization • ternary update • LLM training

TACO: Ternary Absolute-max Column-wise One-sparse Optimizer for LLM Fine-Tuning

174× Less Optimizer State Than AdamW8bit, Full-Parameter Fine-Tuning of 32B Models on One 80 GB H100, Comparable Accuracy

By Jichao Jiang, Cristian McGee, El Houcine Bergou, Hanqin Cai & Aritra Dutta (University of Central Florida; Mohammed VI Polytechnic University)

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TACO takes the operator-norm steepest-descent view to its logical extreme: it selects the sign of the largest-magnitude entry in each column of every two-dimensional weight matrix—a ternary, one-sparse-per-column update that requires essentially zero persistent optimizer state—and reduces AdamW8bit’s 27.7 GB optimizer footprint on OPT-13B to 0.16 GB while matching downstream accuracy across multiple model families and enabling 32B-scale full-parameter fine-tuning on a single H100.

AT A GLANCE

Problem: AdamW and quantized variants require 2–3× the model size in optimizer state, limiting full-parameter fine-tuning to small models on a single GPU.

Why it matters: Full-parameter fine-tuning outperforms LoRA on complex domain adaptation; enabling it for 30B+ models democratizes access.

Method: Exact steepest descent under dimension-normalized 1→1 operator norm selects sign of the maximum-magnitude column entry; no persistent momentum buffers.

Result: 174× optimizer state reduction; 2.9× peak memory reduction; accuracy within 0.5–1.0% of AdamW on SuperGLUE, SQuAD v2.

Evidence: OPT, LLaMA, Mistral, Phi families; fine-tuning tasks including SuperGLUE, SQuAD v2, and instruction following.

The Big Picture

Full-parameter fine-tuning of large language models is the gold standard for domain adaptation: it updates every weight, avoids the rank constraints of parameter-efficient methods (LoRA, prefix tuning), and captures distribution shifts that low-rank approximations cannot

represent. Yet it has been practically out of reach for models above ~7B parameters on a single 80 GB GPU because AdamW—the dominant optimizer—maintains first and second moment buffers that add 2× the model parameter count in optimizer state.

TACO addresses this by rethinking what state an optimizer actually needs. Drawing on the operator-norm steepest-descent perspective introduced by Muon, TACO shows that the exact optimal update under a dimension-normalized 1→1 operator norm requires no momentum buffers whatsoever—only the current-step gradient, which is discarded immediately after use.

Why Existing Approaches Fall Short

AdamW: Two full-precision moment buffers (first and second order) add 2× the model’s FP32 parameter count to GPU memory. On OPT-13B this is 27.7 GB of persistent state alone.

AdamW8bit: Quantizes moment buffers to 8 bits, cutting state to $\approx 27.7 \text{ GB}/2 = 13.9 \text{ GB}$ —a useful reduction but still far larger than TACO’s 0.16 GB.

SignSGD: Takes the sign of the full gradient column-by-column but does not arise from a principled operator-norm steepest-descent argument, and can exhibit slow convergence on poorly scaled gradients.

Muon: Derives updates via operator-norm steepest descent and applies Nesterov momentum, but still maintains a first-moment buffer.

LoRA: Reduces state by fine-tuning only low-rank adapters, but cannot update all parameters and underperforms full-rank methods on complex adaptation.

Core Mechanism

Operator-norm steepest descent. Given a weight matrix $W \in \mathbb{R}^{m \times n}$ with gradient G , the steepest-descent direction under the dimension-normalized $1 \rightarrow 1$ operator norm constraint selects, for each column j , the row index achieving the maximum gradient magnitude, and assigns a ternary entry equal to the sign of that element.

Formally, the update matrix $U \in \{-1, 0, +1\}^{m \times n}$ has exactly one nonzero entry per column—hence “one-sparse per column” and “ternary”. No previous-step information is accumulated: the only state at any step is the current gradient, discarded immediately after U is computed.

Technical Formulation

For each column $j \in [n]$ of gradient $G^{(t)}$:

$$i^*(j) = \arg \max_{i \in [m]} |G_{ij}^{(t)}|$$

$$U_{ij}^{(t)} = \begin{cases} \text{sign}\left(G_{i^*(j),j}^{(t)}\right) & i = i^*(j) \\ 0 & \text{otherwise} \end{cases}$$

Weight update:

$$W^{(t+1)} = W^{(t)} - \eta U^{(t)}$$

Persistent optimizer state between steps: $\emptyset$ (the BF16 gradient buffer $G^{(t)}$ is freed immediately after the update).

Under a convex loss with Lipschitz gradient, TACO achieves $O(1/\sqrt{T})$ convergence matching signSGD, while the update $U^{(t)}$ is the exact minimizer of the linearized objective subject to the unit operator-norm ball.

Learning or Inference Procedure

Forward: Standard.

Backward: Compute gradient $G^{(t)}$ for each 2D weight matrix.

Update (per-matrix):

1. Compute argmax over each column of $|G^{(t)}|$.
2. Construct ternary one-sparse matrix $U^{(t)}$.
3. Apply $W \leftarrow W - \eta U^{(t)}$ in-place.
4. Free $G^{(t)}$.

Non-2D tensors (embeddings, 1D biases) fall back to signSGD. Learning rate: cosine or linear warmup-decay schedule; no additional hyperparameters beyond η .

What the Guarantee Says

For convex losses, TACO converges at rate $O(1/\sqrt{T})$, matching signSGD. The $U^{(t)}$ is the exact steepest-descent direction under the $1 \rightarrow 1$ operator norm: no other ternary update achieves a greater linearized objective decrease under the operator-norm geometry. The persistent state between steps is strictly zero beyond the model weights, providing a hard lower bound on optimizer memory that no further quantization can improve.

Experimental Findings

- **Optimizer state:** 27.7 GB (AdamW8bit) $\rightarrow$ 0.16 GB (TACO) on OPT-13B: a $174\times$ reduction.
- **Peak training memory:** 80.6 GB $\rightarrow$ 27.5 GB on OPT-13B: a $2.9\times$ reduction.
- **Large-model access:** OPT-30B and LLaMA-32B can be fully fine-tuned on a single 80 GB H100—previously impossible with AdamW.
- **Accuracy:** Within 0.5–1.0% of AdamW on SuperGLUE, SQuAD v2, and instruction following, across OPT, LLaMA, Mistral, Phi.
- **Throughput:** Training step runtime within 5% of AdamW8bit owing to the simplicity of the argmax-and-sign update.

Ablations and Interpretation

Row-wise vs. column-wise argmax: Row-wise argmax (violating the operator-norm derivation) slightly degrades accuracy, confirming the theoretical motivation matters in practice.

Top- k per column (TACO- k): Allowing $k > 1$ nonzero entries per column interpolates between TACO and a dense gradient, recovering AdamW accuracy at the cost of k -fold increased state; $k = 1$ is Pareto-optimal for memory-constrained settings.

Momentum ablation: Adding a first-moment moving average of $U^{(t)}$ improves convergence speed on some tasks but doubles persistent state; omitting momentum is preferred when memory is the bottleneck.

TACO vs. LoRA: At equal GPU memory budget, full-parameter TACO outperforms LoRA on domain-specific tasks, confirming the value of full-rank updates when memory constraints are relaxed by the optimizer.

Reference

Jichao Jiang, Cristian McGee, El Houcine Bergou, Hanqin Cai, Aritra Dutta. **TACO: Ternary Absolute-max Column-wise One-sparse Optimizer for LLM Fine-Tuning**. arXiv:2610.02199, October 2026.
<https://arxiv.org/abs/2610.02199>